

110 MILES THE DIAMETER OF THE CHICXULUB CRATER

This radar image shows a small portion of the Chicxulub crater, Mexico. The crater was first discovered nearly 20 years before geologists realized it was caused by the object that probably contributed to the demise of the dinosaurs.

THE INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE

(see 1988) published its first assessment report in 1990. The report suggested that the average **global temperature was increasing** by about 32°F (0.3°C) per year due largely to man-made emissions of greenhouse gases, such as carbon dioxide (see pp.326–27).

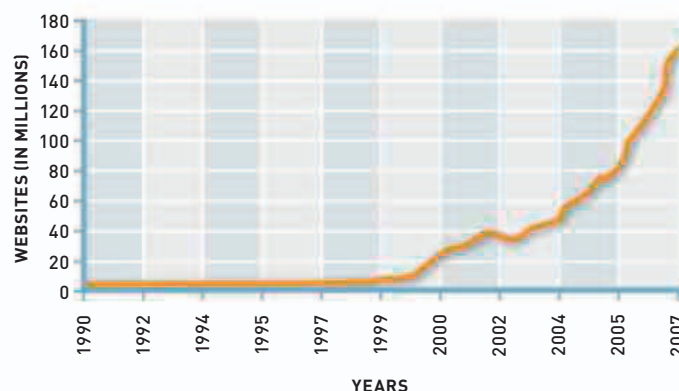


TIM BERNERS-LEE
(b.1955)

British computer scientist Tim Berners-Lee earned a physics degree from Oxford University. While working at the European Organization for Nuclear Research (CERN) he developed the concept of the Web. In 1994, Berners-Lee founded the World Wide Web Consortium (W3C), the international organization that develops Web standards.

It set out some of the **potential impacts of global warming**, such as rising sea levels and threats to biodiversity. Further assessment reports have maintained and refined the scientific conclusions.

British computer scientist **Tim Berners-Lee** created the **world's first Web browser**—called WorldWideWeb. At the time, the internet was growing rapidly, but was mainly used by academics, communicating via typed commands on bulletin boards—systems that allowed users to exchange software and post messages. There were several different operating systems in use, and few common programs or document formats. Berners-Lee devised a computer language—**hypertext markup language (html)**—that could be used on any computer, to create pages of information. Crucially, these pages could contain links to pages on other, specially programmed, internet-linked computers called servers. The result would be a “web” of information—hence the name of the software, and eventually the World Wide Web itself. Berners-Lee created the first web server at CERN, in Switzerland, where he was working at the time.



Rise in number of websites

The number of websites rose steeply after the internet became commonplace among businesses and home users.

could differentiate between oxygenated and deoxygenated blood. Ogawa realized that this could **reveal which regions of a brain are most active**. His technique forms the basis of functional MRI (fMRI), which is used to measure brain activity. He produced fMRI images of rats in 1990; the first human fMRI imaging was carried out in 1992.

In June 1990, researchers in the US began the world's first clinical **trial to use gene therapy**. A gene was inserted into white blood cells, and the resulting transgenic cells were injected into a girl suffering from a severe immune disorder.

In April 1990, NASA's Space Shuttle *Discovery* carried the **Hubble Space Telescope (HST)** into a low-Earth orbit. The telescope, named after the American astronomer Edwin Hubble (see 1923), carries a range of instruments that can detect infrared, ultraviolet, and visible light. The Hubble

Space Telescope has produced stunning images of a wide range of objects in space, and has provided huge amounts of information for astronomers, astrophysicists, and cosmologists.

Japanese biological physicist **Seiji Ogawa** (b.1934) developed an extension of **magnetic resonance imaging (MRI)** that



Hydrogen fueled car

In 1991, the Japanese firm Mazda revealed its first HR-X concept car, which has an internal combustion engine that burns hydrogen.

February 14, 1990 The Voyager 1 space probe turns around to take the “Family Portrait” of all the planets except Mercury and Mars

April 24, 1990 Hubble Space Telescope launched

April 1990 Launch of the Human Genome Project

June 1990 US company General Instruments introduces first digital high-definition television system

September 14, 1990 The first gene therapy carried out

September 10, 1990 The first internet search engine, called Archie, released

September 15, 1990 NASA's Magellan spacecraft began detailed 3-D mapping of the surface of Venus

October 26, 1990 The IPCC published its First Assessment Report on climate change

December 1990 Technology underlying fMRI, which allows brain processes to be imaged in real time, is developed

December 25, 1990 The world's first web browser, called WorldWideWeb, is created